PROJECT REPORT · 4 NOVEMBER 2026

KrugWorld

A gamified life app styled as an RPG character sheet, with an AI steward who also lives on your desk.

25 days, 8 Oct to 2 Nov	**41** commits	**~21k** lines of own code	**143** backend tests, all green
85 API endpoints	**14** Flyway migrations	**29** tools for Gaby	**1** live site

Live at krugworld.veteryr.online · Source: github.com/Morker/krugworld (private)

Every number in this report was measured from the repository and the live site on 4 November 2026.

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How to read the charts

Bars are sorted longest first. Blue is the main measure. Orange marks a second kind (unit tests, writes, the device API) and gray marks code that wasn't written for this project. Each chart has its numbers printed on it, so nothing depends on colour alone.

SECTION 01

Summary

KrugWorld turns everyday life admin into an RPG character sheet. Money, plans, meals, training and ideas all earn XP. Gaby, an AI steward, can read and change all of it, in the browser or out loud on a desk device.

~21k	143	85	23
lines of own code	backend tests	API endpoints	database tables
14	29	12	2
migrations, all applied live	Gaby tools	badges to earn	clients: PWA + ESP32

Where it stands

- **Live** since 15 October 2026 at krugworld.veteryr.online, behind nginx with Let's Encrypt HTTPS, as one Docker container next to its own Postgres.
- **Feature-complete against the old Krugger's Tavern**, plus everything the Tavern never had: Gaby, Desk Gaby, phone notifications, debts, password reset and an admin desk.
- **Latest release, 2 November 2026:** schema V11 to V14 (recurring seal, debts, admin). Confirmed on the live status endpoint, which reports schema version 14.
- **Green build:** 143 of 143 tests pass; the frontend lints at zero warnings; CI runs both on every push to main.

What stands out

- **Money can't drift.** Balances and debts are always summed from history in SQL, never stored, so there is no second copy to fall out of step.
- **Rewards are paid once.** Every XP payment goes through a unique (user, kind, ref) key with `INSERT … ON CONFLICT DO NOTHING`: retries and double taps can't farm it.
- **One assistant, not two.** Gaby's 29 tools call the same services the screens use, so every rule and check applies to her too.
- **Tested against the real database.** 111 of the 143 tests boot the whole app on a throwaway Postgres in Docker (Testcontainers), not an in-memory stand-in.

SECTION 02

Features

Five groups in the menu, the same on desktop and phone (where each group is a tab in the bottom bar).

Group	Screen	What it does
Hall	Character sheet	Balance as an HP bar, budgets as mana, quests, streak flames, badges, the latest loot, Gaby's line of the day
Finance	Ledger	Every coin in and out, by month, account and category; transfers between accounts
	Budgets	Monthly limit per category, read against real spending; +50 XP for a month kept
	Quests	Savings goals with progress; +200 XP when one completes
	Recurring	Salary, rent, bills: book themselves nightly, or wait for your seal so a varying bill gets its real amount
	Debts	Who owes you, whom you owe; tick off when settled (+10 XP), reminders before and on the due day
	Accounts	Bank, cash, e-wallet, card; archiving keeps history adding up
Life	Calendar	Events and plans
	Vitals	Meals against a calorie goal, training plan with daily ticks
	Workshop	Ideas from spark to finished (+15 XP)
Arcane	Steward	Chat with Gaby in English or Malay; she acts through tools
	Oracle's Eye	A look across the data
Settings	Settings	Skin and background, recovery email, weather place, phone notifications, Desk Gaby pairing, Tavern import
Admin	Keeper's Desk	Admins only: correct another account's entries or balance; every change audited

Desk Gaby

A Waveshare ESP32-S3 with a 4-inch touch screen, microphone and speaker. It pairs with a code shown in Settings and gets its own revocable token (never the user's password). It shows a glance screen with a blinking Gaby and today's numbers, ticks workouts with a tap, and answers out loud: tap to talk, and her voice streams back with an English subtitle. It talks HTTPS to the live site with the Let's Encrypt roots pinned in the firmware.

Two looks

Classic Krugger keeps the old Tavern exactly (Press Start 2P, VT323, teal stone, scanlines). The new skin uses Pirata One and Barlow Condensed over four backgrounds, with Knight's steel as the default. The skin is a set of CSS variables on the page root; components only use the variables.

SECTION 03

Architecture

One jar serves both the React app and the API, so there's one thing to build, ship and roll back.

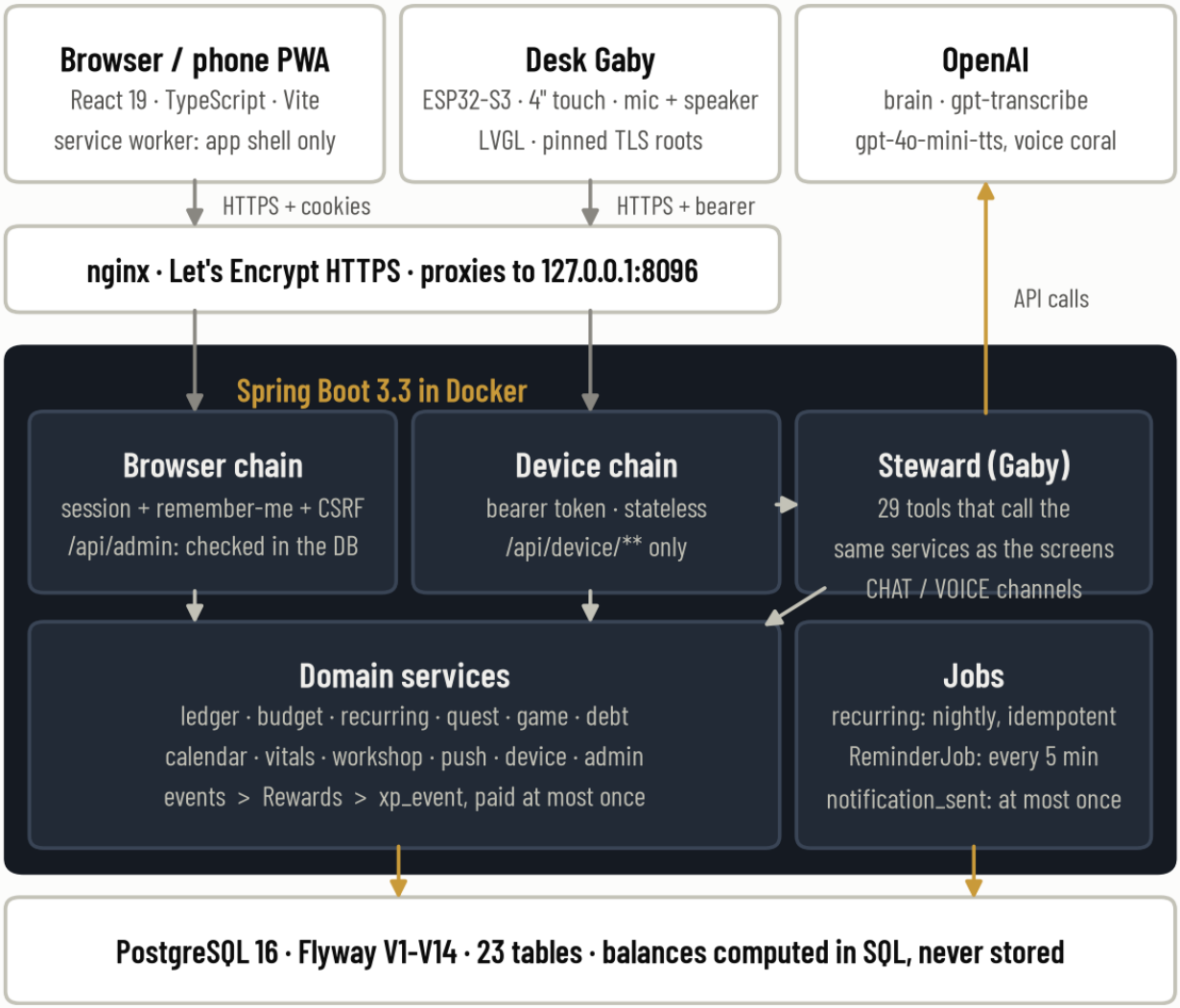


Figure 1. The whole system. Two clients, one server with two separate security chains, one database. Gold arrows leave the server.

Two doors, kept apart

The browser and the desk device never share a way in. `/api/device/**` accepts only a bearer token and keeps no session; everything else is the browser chain with a session, a year-long remember-me cookie and CSRF protection. A device token is ignored on the browser side, and a browser cookie is ignored on the device side.

Admin is checked in the database on every call

The admin flag can only be set with SQL on the server, and the app reads it fresh on each `/api/admin/**` request instead of trusting the session. Granting or revoking takes effect on the next request, with no new login.

SECTION 04

The codebase

About 21,000 lines written for this project, plus documentation and some vendored audio-codec drivers.

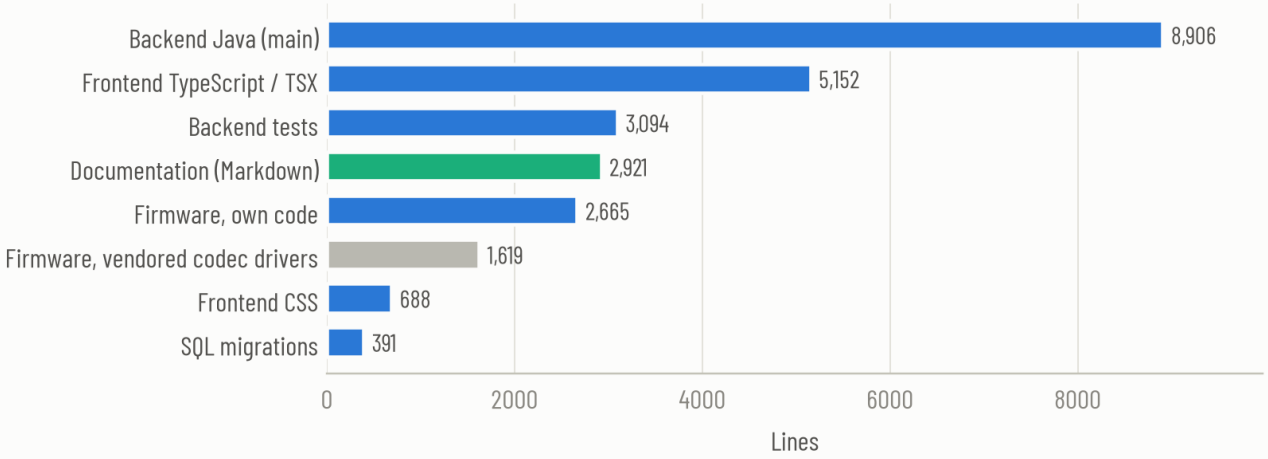


Figure 2. Lines by area. Gray is Espressif's ES8311/ES7210 codec drivers, carried in the firmware but not written here. Aqua is documentation. Lock files and the earlier PDF are left out.

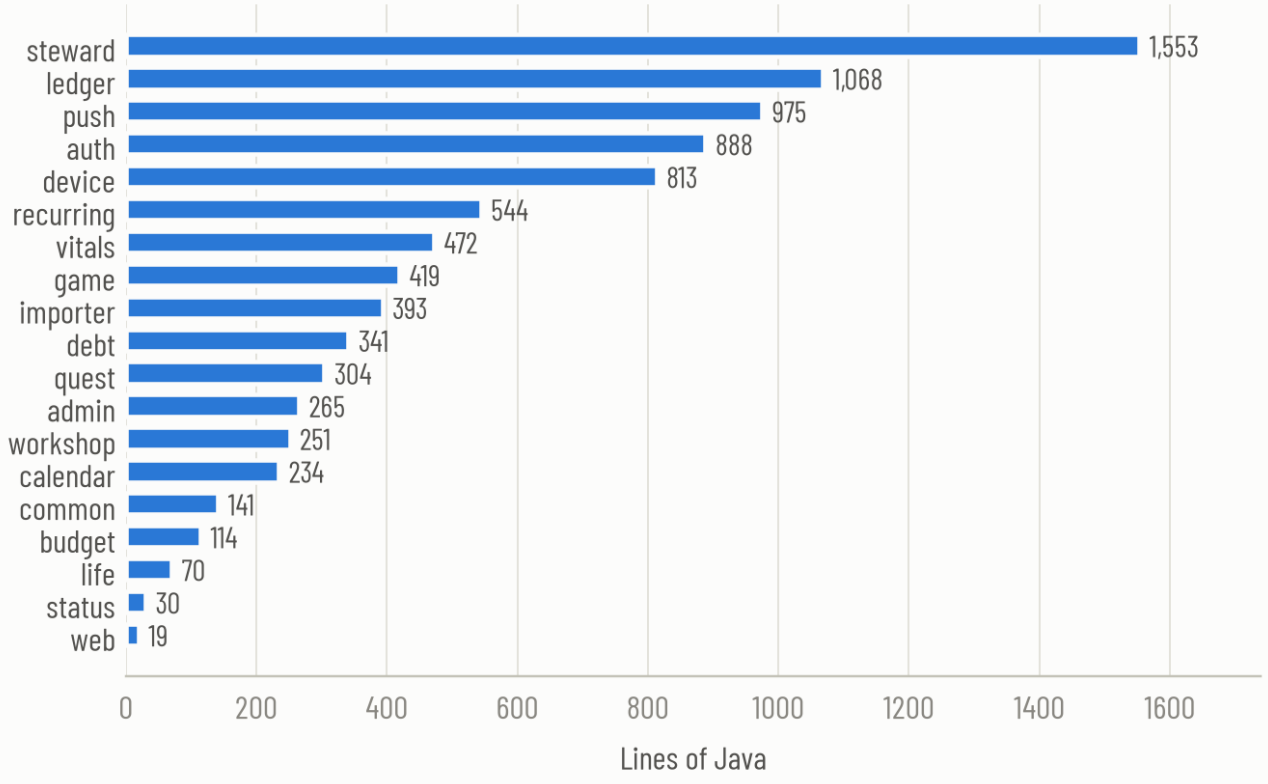


Figure 3. Backend Java by package. The Steward is the largest: its tools, two AI providers and the voice pipeline. Ledger is next, as the heart of the money side.

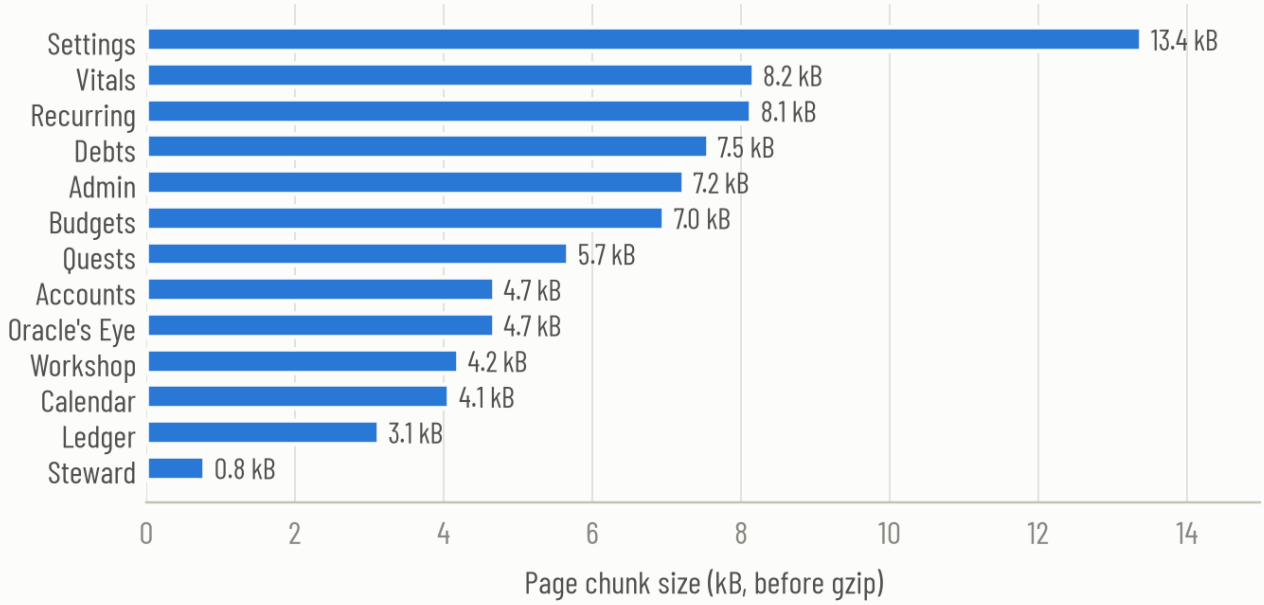


Figure 4. Every page except the Hall is its own lazy-loaded chunk, so a phone only fetches a page the first time it opens it. The shared app bundle is 326 kB, or 101 kB gzipped.

The frontend in brief

- **React 19 + TypeScript + Vite 8**, with React Query caching every GET under a key; a change invalidates exactly the keys it affects (logging an entry refreshes balances, budgets, the month summary and the character sheet).
- **Optimistic updates** for the taps that must feel instant (ticking a workout, deleting an entry), with a rollback if the server says no.
- **Installable PWA.** The service worker caches the app shell only and never `/api`, so money on screen is never a stale copy. Icons are drawn from Gaby's portrait.
- **One page per error boundary.** A crash in one page shows a friendly panel there; the rest of the app keeps working.

SECTION 05

How it grew

Most of the foundation landed on the first day. After that, each step was one feature, built and tested end to end.

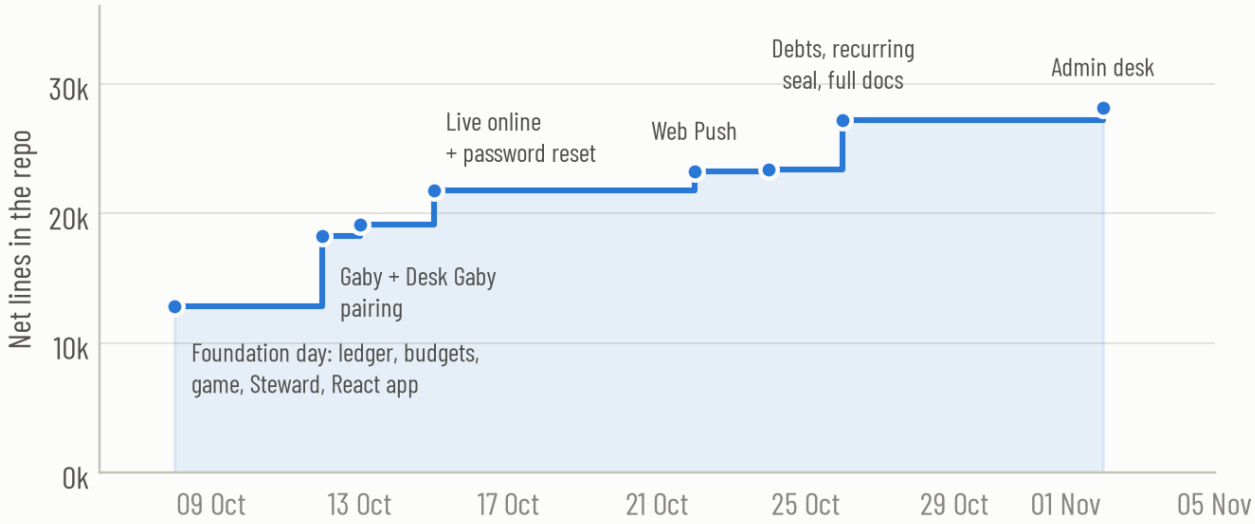


Figure 5. Net lines in the repository after each working day (additions minus deletions; lock files and PDFs left out).

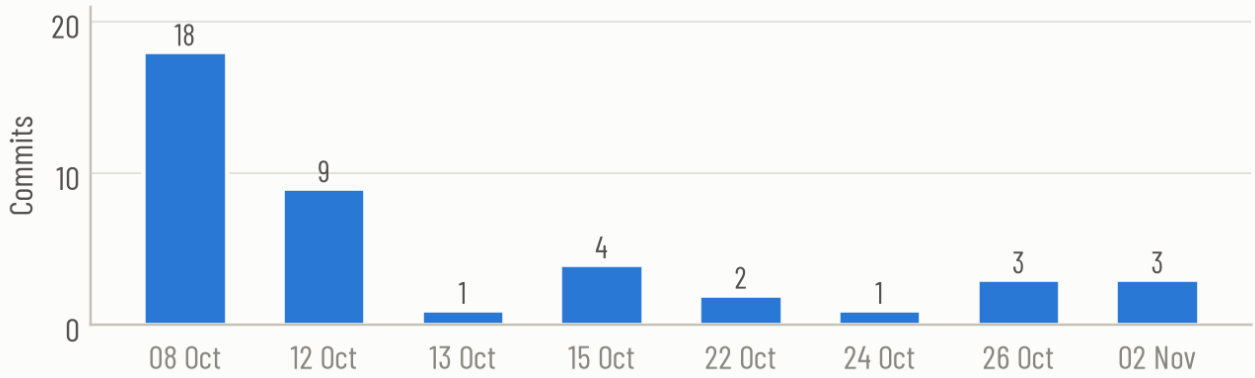


Figure 6. Commits per working day: 41 in total.

Date	Milestone
8 Oct	Spring Boot + Flyway + React scaffold; login with remember-me; ledger, budgets, recurring, game layer, Tavern import, Life pages, the Steward (then called Barnaby); renamed from Coffold to KrugWorld
12 Oct	OpenAI becomes the default brain; Barnaby becomes Gaby (English and Malay); desk device pairing; firmware v1
15 Oct	Online at krugworld.veteryr.online; Desk Gaby on HTTPS; password reset by email
22 Oct	Web Push notifications with hand-written RFC 8291 encryption
24 Oct	Meal and money check-in reminders; browser chat replies in one language
26 Oct	Debts; recurring entries that wait for your seal; the full documentation set
2 Nov	Admin desk with an audit trail; migration collision fixed; V11 to V14 released

SECTION 06

Data

PostgreSQL 16, changed only by Flyway migrations. Hibernate validates the schema but never edits it.

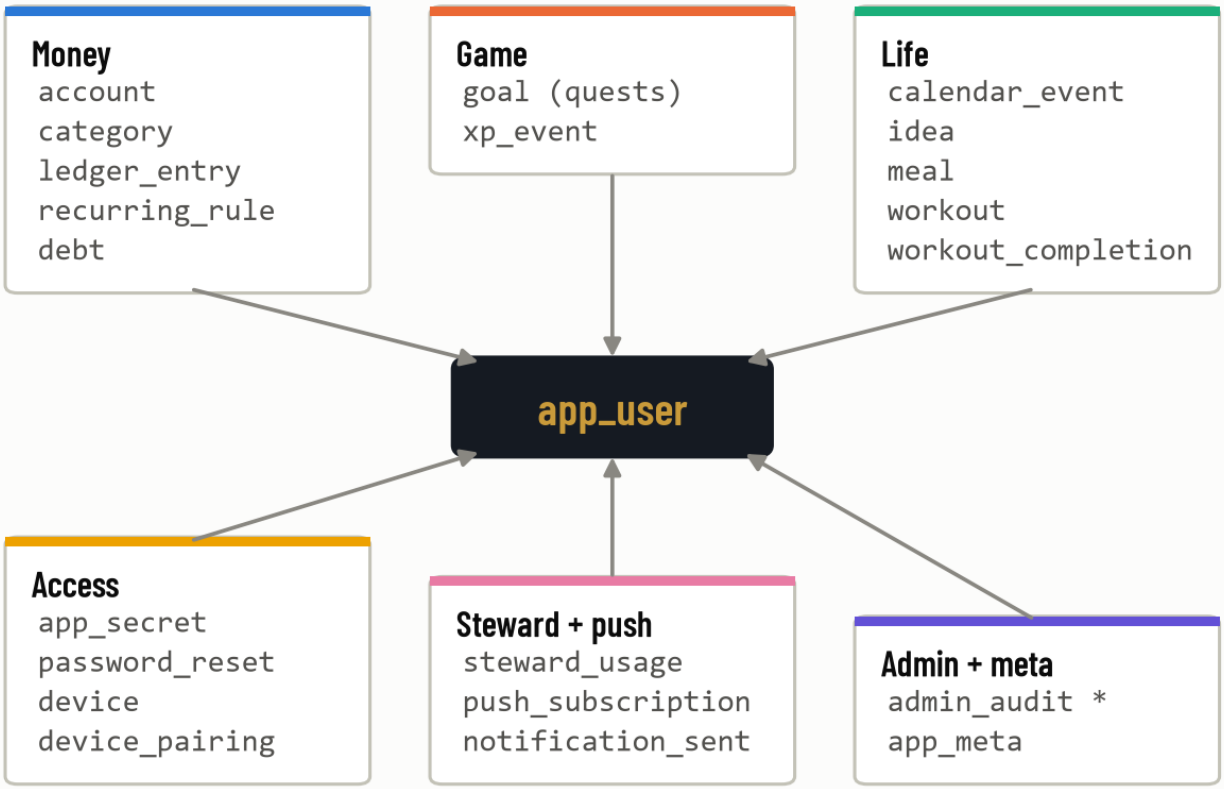


Figure 7. All 23 tables grouped by domain. Every personal table points at app_user with ON DELETE CASCADE. * admin_audit is the exception: it has no cascade, so an admin's record can't vanish by accident.

Rules that keep the numbers honest

- Money is `NUMERIC(14,2)` in the database and `BigDecimal` in Java; the browser sends amounts as two-decimal strings, so no floating-point rounding reaches a balance.
- Every lookup by id also takes the user id, so another user's row looks exactly like a missing one.
- A migration that has run is never edited: a change is always a new `V<n>` file.

Fourteen migrations

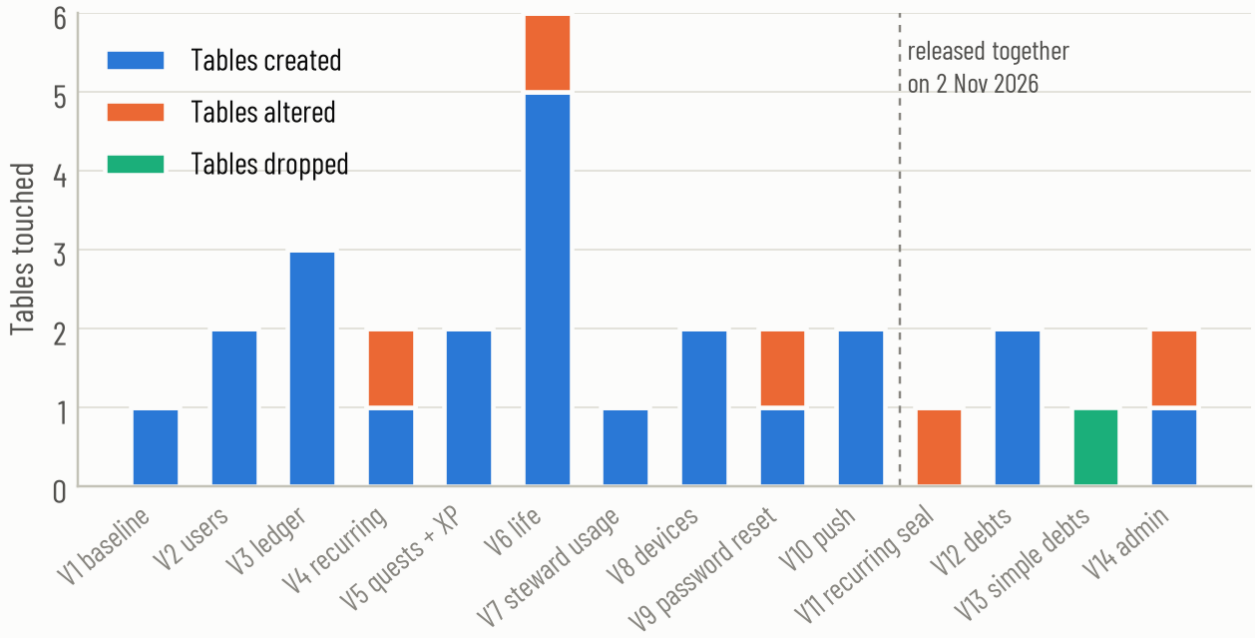


Figure 8. What each migration changed. V13 dropped the repayments table V12 had added, without editing V12: migrations only move forward.

Version	Adds
V1	`app_meta`
V2	`app_user`, `app_secret`
V3	`account`, `category`, `ledger_entry`
V4	`recurring_rule`, `ledger_entry.recurring_rule_id`
V5	`goal`, `xp_event`
V6	`calendar_event`, `idea`, `meal`, `workout`, `workout_completion`, user settings
V7	`steward_usage`
V8	`device`, `device_pairing`
V9	`app_user.email`, `password_reset`
V10	`push_subscription`, `notification_sent`
V11	`recurring_rule.auto_post`: recurring entries can wait for the user's seal
V12	`debt`, `debt_payment`
V13	drops `debt_payment`: debts are ticked off, not repaid in parts
V14	`app_user.is_admin`, `admin_audit`

SECTION 07

The API

85 endpoints across 19 controllers, all speaking JSON. Every error has one shape: `{code, message[, fields]}`.

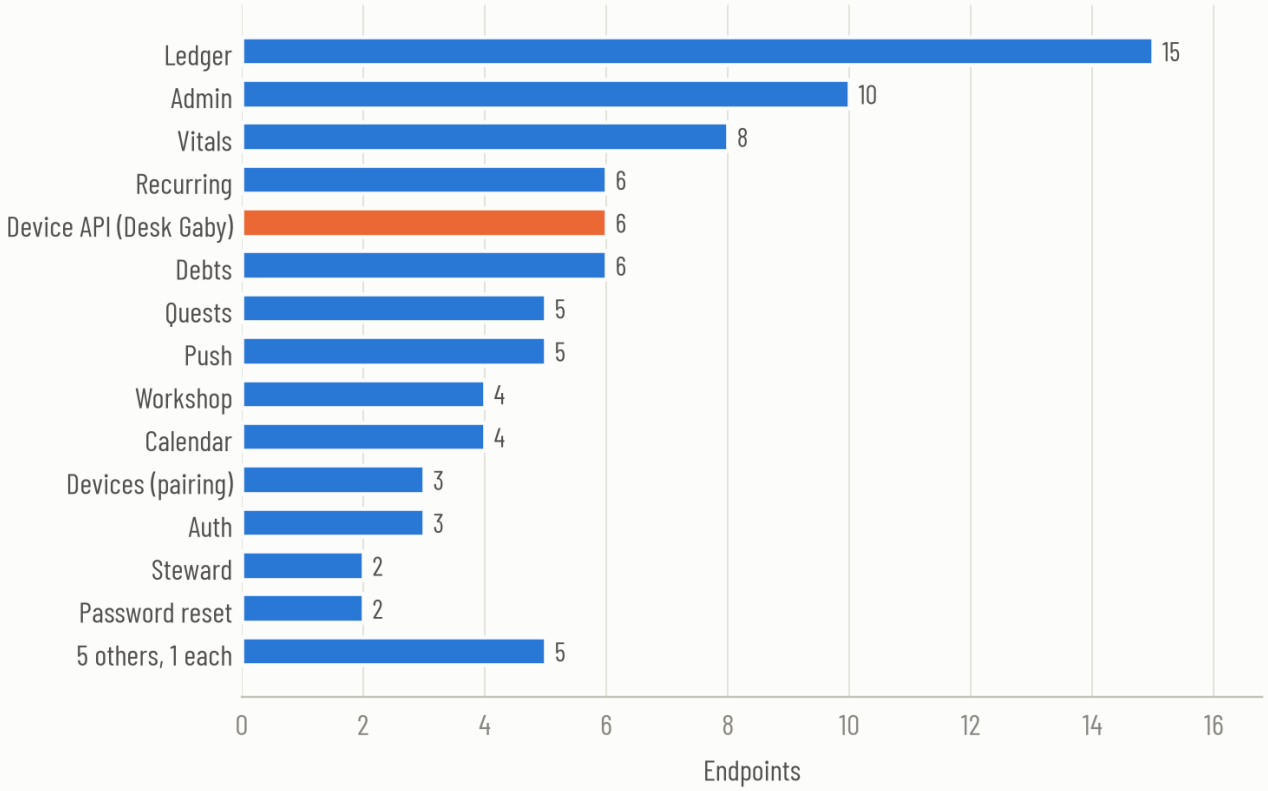


Figure 9. Endpoints per controller. Orange is the device API that Desk Gaby uses with its bearer token; everything else is the browser API.



Figure 10. Endpoints by HTTP method: writes (POST, PUT, PATCH, DELETE) slightly outnumber reads.

Not yet in API.md

The ten admin endpoints under `/api/admin/users` are documented in DEPLOY.md and CLAUDE.md but not yet in docs/API.md.

SECTION 08

Quality and testing

143 backend tests, all passing on a clean build on 2 November 2026. There are 0.35 lines of test for every line of backend code.

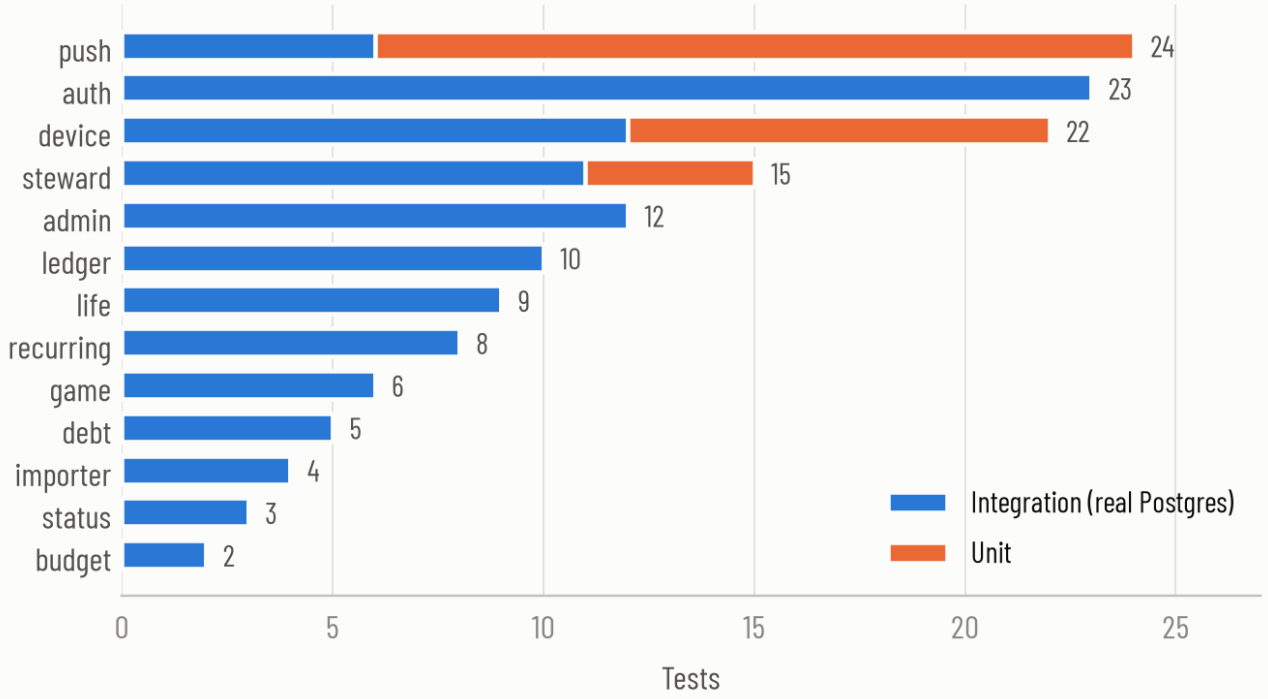


Figure 11. Tests by area. Blue tests boot the whole app on a real Postgres in Docker; orange unit tests cover pure logic (reminder rules, push encryption, Gaby's daily line, the Claude adapter).

Practice	How it's done
Real database in tests	Testcontainers 1.21.4 (pinned: older versions fail against Docker 29); one shared Postgres per run
No network, no cost	Gaby runs on a scripted fake chat and speech provider; mail and push go to fakes
Scheduler off in tests	Jobs are called directly, so no test races a cron trigger
Continuous integration	GitHub Actions on every push to main: frontend lint and build, backend package with all tests
Frontend hygiene	oxlint at zero warnings; TypeScript strict build before every release
Crypto cross-checked	Web Push encryption checked against Node's own crypto implementation

SECTION 09

Gaby, the steward

A pixel-art steward in a burgundy cap with a gold feather. She understands English and Malay, keeps a light medieval flavour, and acts on your data only through tools.

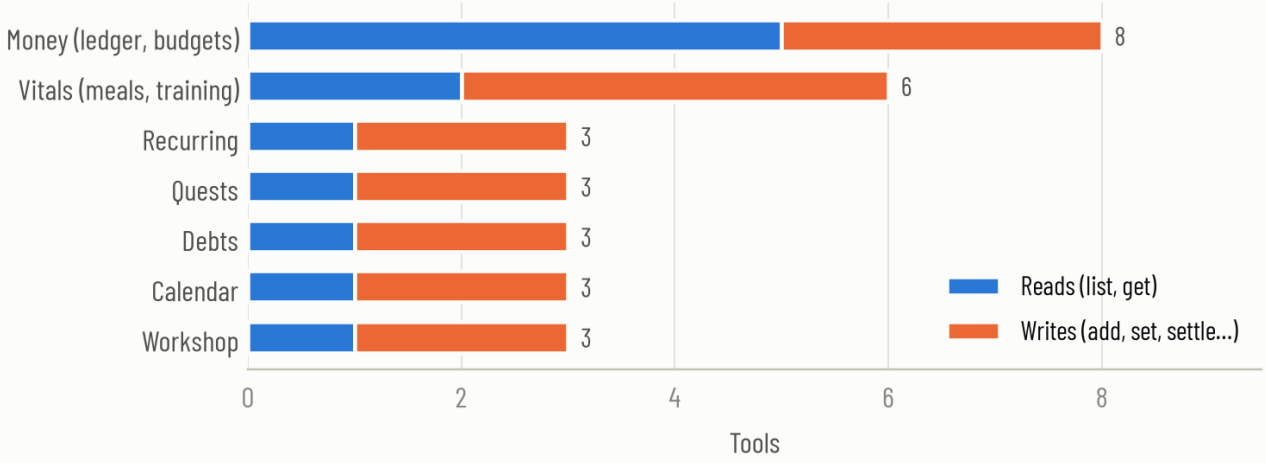


Figure 12. Gaby's 29 tools by domain. 12 read, 17 change something; each one goes through the same service, checks and XP rules as the matching screen.

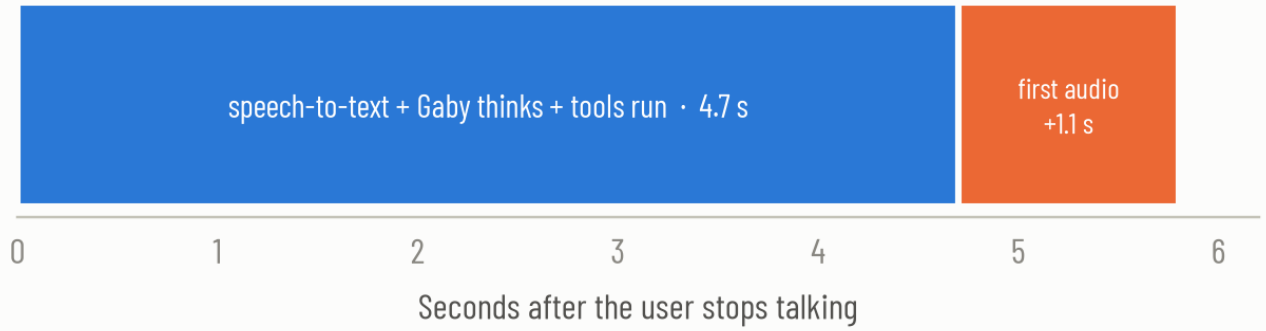


Figure 13. A spoken question on Desk Gaby, measured end to end with real OpenAI on 12 October 2026. The voice streams, so she starts talking 1.1 s after her text reply is ready.

Where	Language rule
Browser chat	One language per reply: whichever the user's message is mostly in, never mixed
Desk Gaby (spoken)	Keeps the rojak mix of English and Malay, the way people actually talk
Screen text	Always English: device labels, her daily line, and the subtitle under a spoken reply
Providers	OpenAI for brain, speech-to-text and voice (coral); Claude available by setting one variable

SECTION 10

The game layer

XP rewards habits, not volume: the cap on entries and the once-a-day rules make farming pointless.

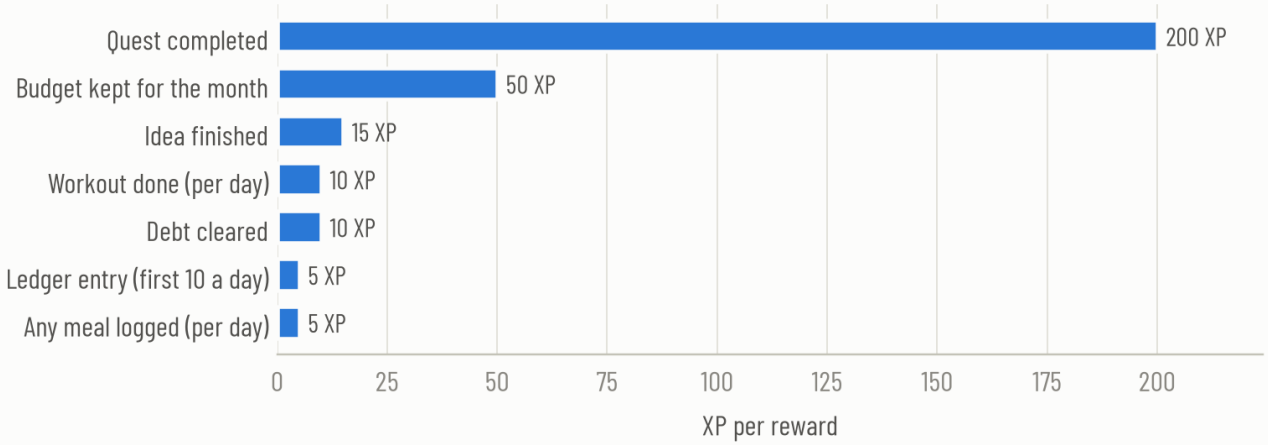


Figure 14. What each action pays. Entries earn XP for the first ten a day only; meals and workouts pay once per day.

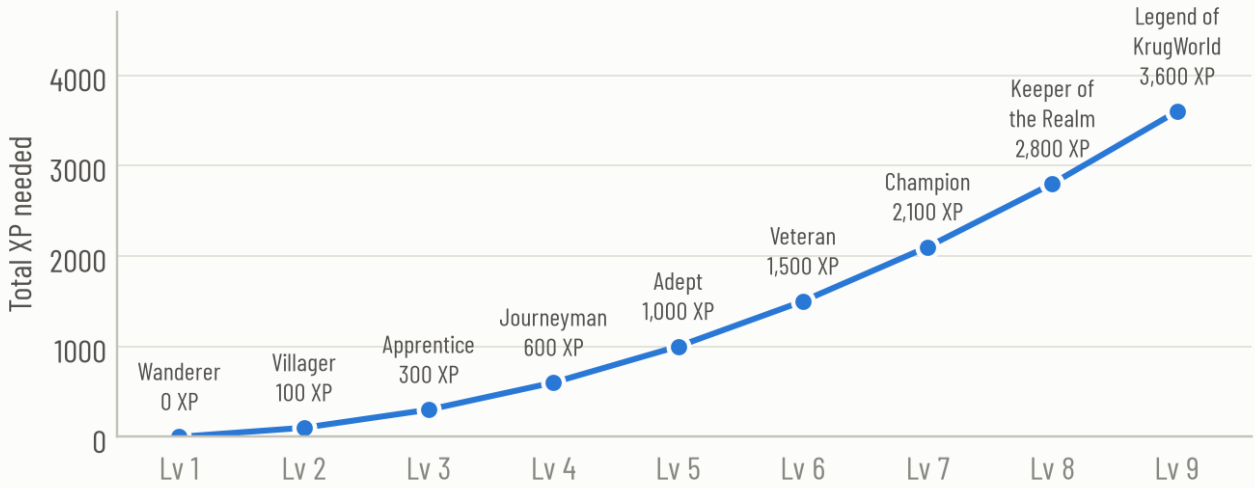


Figure 15. Level n needs 100 × (n − 1) more XP than the level before, so early levels come quickly and later ones take real habits. Nine titles, from Wanderer to Legend of KrugWorld.

12 badges

First coin · Week warrior · Iron habit · Budget keeper · Quest complete · No-spend Sunday · Hoard of a thousand · First feast · Iron body · Idea smith · Clean slate · Adept

SECTION 11

Operations

A small, shared server, and a release process built from two real mistakes.

Piece	Setup
Server	Tencent Lighthouse box shared with other projects; KrugWorld lives in /opt/krugworld and listens on 127.0.0.1:8096 only
Containers	The app and its own postgres:16, with docker compose
HTTPS	Host nginx with Let's Encrypt, auto-renewing; buffering is off for Gaby's voice route so her speech streams
Release	build.ps1 (clean build + tests) on the PC, upload the jar, redeploy.sh; Flyway migrates on start
Secrets	Server .env holds the database password and the OpenAI key; remember-me and VAPID keys live in app_secret
Health check	/api/status reports the schema version, which is how the 2 November release was confirmed

Two migration collisions, one lesson

When	What happened	Fix
23 Oct	Two PCs each created a V10. The release from one PC met the other's V10 on the server, Flyway refused the checksum, and the site was down (no data touched).	Migrations renumbered; one combined release from pushed code
2 Nov	A merge brought two different V11 files. Git saw no conflict because the names differed; the app would not have started. Found before release.	Admin renumbered to V14; build.ps1 now cleans first so a stale copy can't ride along in the jar

The rule now in the project briefing

Release only code that is pushed to GitHub, and number new migrations above production's latest version. Check production with `/api/status` before picking a number.

SECTION 12

Open items and next steps

Item	Why it matters	Suggested next step
Push not tried on a real phone	Notifications were built and tested against fakes and Node's crypto, not yet on a device	Turn it on in Settings on an Android phone and send a test
Admin endpoints missing from API.md	The API reference is the one place a reader expects to find them	Add the ten /api/admin/users endpoints
Status line in CLAUDE.md is dated 12 Oct	It still lists 73 tests; the code has 143	Refresh the status block
Releases are built by hand	Both incidents came from a jar built on a PC that wasn't in step with GitHub	Let GitHub Actions build and deploy after the tests pass
Admin rides on a year-long login	An admin session lasts as long as any remember-me cookie	Consider asking for the password again before admin changes
Desk Gaby roadmap	Quick-log buttons, device reminders and moods, then a 'Hey Gaby' wake word	Reuse Reminders.due for device nudges; try ESP-SR or microWakeWord

Sources

Line counts, packages, tests, endpoints and migrations were counted from the main branch at commit 5aefc0d. Growth and commits come from git history. The voice timing is the end-to-end test of 12 October 2026. The live schema version was read from /api/status after the 2 November release.